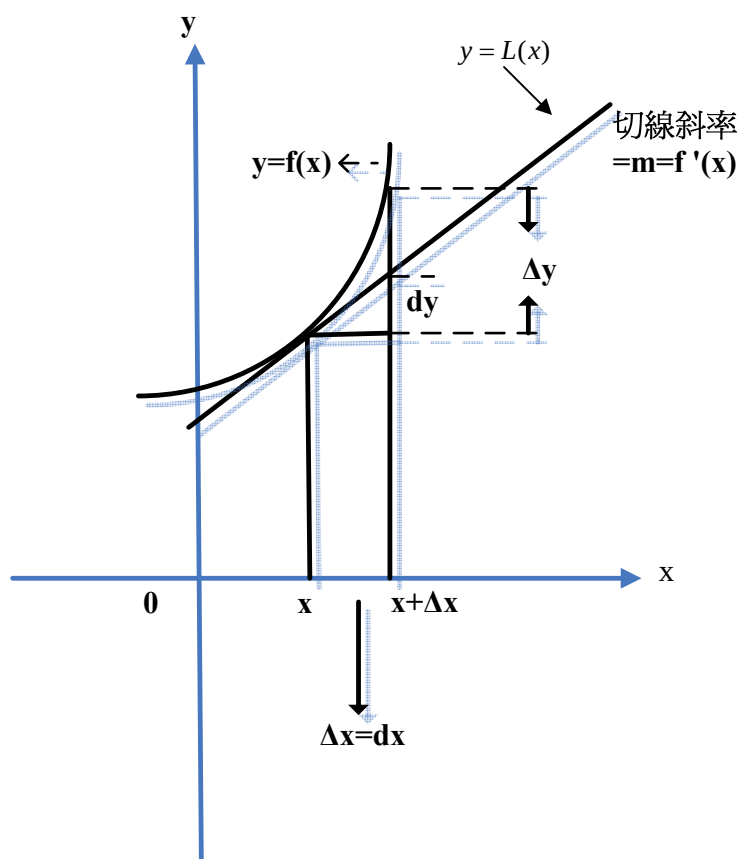


§3-11 Linear Approximations and Differentials

Homework : 7,17,23,31,39,43

給 $f(x)$, 利用過 a 點的切線方程式來逼近當 x 靠近 a 的 $f(x)$.



- $\Delta x = dx$ = initial error
- Approximate error = $dy = f'(x) dx$
- Exact error = $\Delta y = f(x + \Delta x) - f(x)$
- dx and dy are called differentials

公式

1. $\Delta y \approx dy$ when $\Delta x = dx$ is small.

在實際問題，我們常能以 dy (較易算) 來逼近 Δy (較難算).

2. $L(x) = y = f(a) + f'(a)(x - a)$ is the linear approximation or tangent line approximation of f at a .

- $y = L(x)$ 是過 $(a, f(a))$ 和 $f(x)$ 相切的切線方程式.
- $\lim_{x \rightarrow a} (f(x) - L(x)) = 0$ ($\Rightarrow$ 誤差 $= |f(x) - L(x)|$ 趨近於 0, 當 $x \rightarrow a$).
- $\lim_{x \rightarrow a} \frac{f(x) - L(x)}{x - a} = 0$ ($\Rightarrow$ 誤差趨近於 0 的速度比 $x \rightarrow a$ 趨近於 0 的速度還快).

Example1 :

(i) Find the linearization of

$$f(x) = \sqrt{x+3} \quad \text{at } a=1.$$

(ii) Use it to approximate $\sqrt{3.98}$ and $\sqrt{4.05}$.

Solution :

$$\begin{aligned} f'(x) &= \frac{1}{2}(x+3)^{-\frac{1}{2}} \Rightarrow f'(1) = \frac{1}{4} \\ L(x) &= 2 + \frac{1}{4}(x-1) \\ \Rightarrow \sqrt{3.98} &\approx 2 + \frac{1}{4}(0.98-1) = 1.995 \\ \sqrt{4.05} &\approx 2 + \frac{1}{4}(1.05-1) = 2.0125. \end{aligned}$$

Example2 :

The radius of a sphere was measured and found to be 21 cm with a possible error in measurement of at most 0.05 cm. What is the maximum error in using the value of radius to compute the volume of the sphere? (Use differential to estimate such maximum error.)

Solution :

$$\begin{aligned} V &= \frac{4}{3}\pi r^3 \\ \Rightarrow \frac{dV}{dr} &= 4\pi r^2 \Rightarrow dV = 4\pi r^2 dr \\ &= 4\pi(21)^2(0.05) \\ &\approx 277. \end{aligned}$$

Example3 :

Approximate $\ln(1.05)$.

Solution :

$$\begin{aligned}y &= \ln x \\ \Rightarrow dy &= \frac{dx}{x} = \frac{0.05}{1} = 0.05 \\ \Rightarrow \ln(1.05) &\approx \ln 1 + 0.05 = 0.05.\end{aligned}$$